Dr Infrared Heater Remote Thermostat

Node-RED Flows Explained

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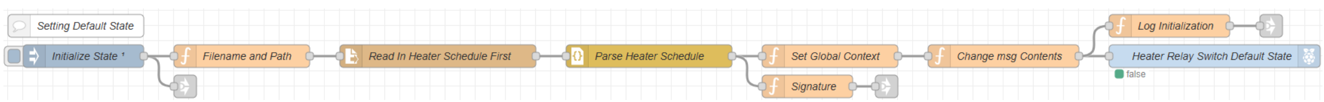
Basics

Node-RED has multiple tabs in this case there are Main, Heater Logic, Min Room Temp, and Debug. Each tab has flows. The flow is multiple nodes connected via wires. Node-RED has three levels of context node, flow and global. Node context is where only a node knows that information for itself. Flow context is where information can be seen by all connected nodes or in the same tab. Global context is where information can be seen by any node. Node names that are highlighted blue have default values that you can easily change such as default minimum room temperature and more. Node-RED uses messages (msg) to pass information between nodes via the wires. The most common message is msg.payload.

Main Flows

Setting Default State



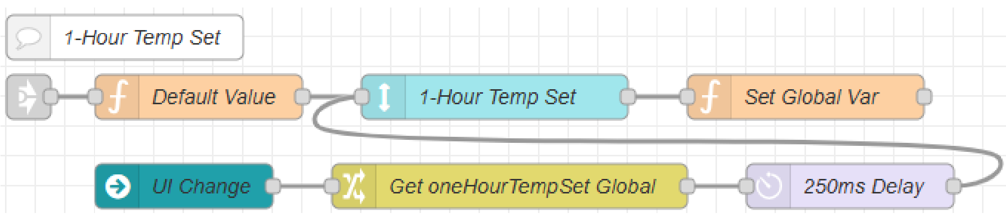
Overview – This flow runs 100ms after Node-RED boots and initializes variable default values and logs status.

Node Details

- Initialize State
  - Injects at 100ms after boot to start flow.
- Filename and Path
  - Sets filename path for HeaterSchedule.json.
- Min Room Temp Out (White arrow block)
  - Connects to "1-Hour Temp Set In" and "Min Room Temp In".
- Read In Heater Schedule First
  - Reads in HeaterSchedule.json file if it exists.
- Parse Heater Schedule
  - Parses heater schedule from JSON (JavaScript object notation) to JavaScript object.
- Set Global Context
  - Set variables heaterSchedule, currentHeaterPinState, manualControlState, oneHourSwitchState and tempProbeError to default values.

- Signature
    - Sets msg.signature to “init” letting the following node know where the message came from.
- Init out (White arrow block)
    - Connects to “Init in 1” and “Init in 2” nodes.
- Change msg Contents
    - Change msg.payload to false.
- Log Initialization
    - Creates log message indicating that the system has booted.
- Log Initialization Out (White arrow block)
    - Connects to “Write to Heater Log In” node to write to HeaterState.log.
- Heater Relay Switch Default State
    - Sets relay pin state low when control system boots.

1-Hour Temp Set



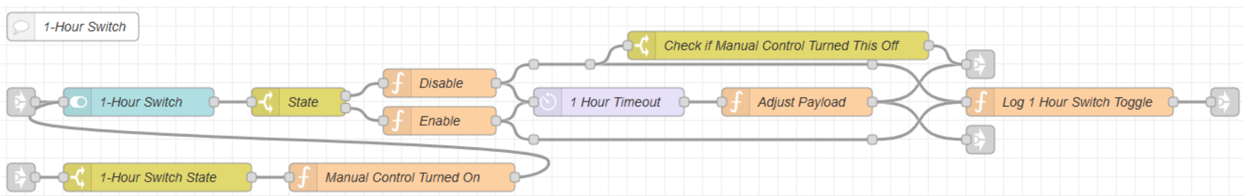
Overview – This flow sets the default valve of the 1-Hour Temp Set and updates a global variable.

Node Details

- 1-Hour Temp Set In (White arrow block)
    - Gets message from “Min Room Temp Out” and routes to next node.
- Default Value
    - Holds integer for default value of 1-Hour Temp Set which is 60°F. You can change this by selecting the node and changing the value of 60 to your desired temperature then deploy your change. I would not go higher than what you are comfortable heating your room too or make the heater run too often.
- 1-Hour Temp Set
    - Numeric user interface (ui) node that allows the user to change this value from the ui. The default range is 0°F to 80°F to change this select the node then deploy your change.

- Set Global Var
    - Sets global variable “oneHourTempSet” when changed from ui.
- UI Change
    - Detects a refresh or page change happens to make sure that the oneHourTempSet value displayed is accurate.
- Get oneHourTempSet Global
    - Get the stored value for the oneHourTempSet.
- 250ms Delay
    - Delays message by 250ms.

1-Hour Switch



Overview – This flow monitors the state of the 1-Hour Switch and will direct the proper information to the heater logic to turn the heater on or off. This also automatically turns the 1-Hour switch off after an hour.

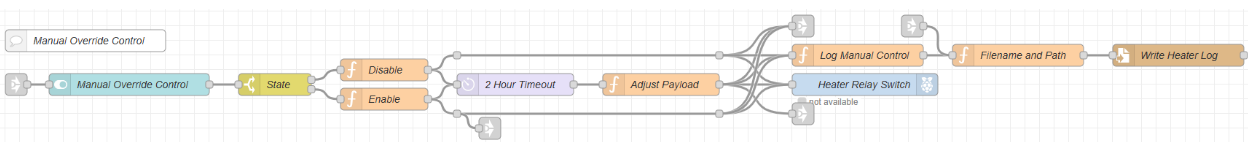
Node Details

- 1-Hour Switch Control In (White arrow block)
    - Receives message from 1-Hour Switch Control Out.
- 1-Hour Switch
    - UI switch for 1-hour switch.
- State
    - Check if 1-hour switch has been turned on or off.
- Disable
    - Sets “oneHourSwitchState” global variable to false, msg.payload to false, and msg.reset to blank to reset the 1-hour timer countdown to shutoff the heater.
- Enable
    - Sets “oneHourSwitchState” global variable to true and msg.payload to true.
- 1 Hour Timeout
    - Tracks a 1-hour timer to shut off switch after an hour.
- Check if Manual Control Turned This Off

- If manual control override switch was turned on when the 1-hour switch is on, then this checks that to update the heater indicator light.

- Adjust Payload
    - Sets msg.payload to false to turn the 1-hour heater switch off after an hour.
- LED Indicator 1-Hour Switch Control Out (White arrow block)
    - Links to ui LED indicator to update.
- Log 1 Hour Switch Toggle
    - Creates log message when the 1-hour switch is turned on or off.
- 1-Hour Switch Control Out (White arrow block)
    - Passes 1-hour timeout shut off message to “1-Hour Switch Control In” to update ui.
- Log 1 Hour Switch Out (White arrow block)
    - Sends log message to “Write To Heater Log In” to write to HeaterState.log.
- 1-Hour Switch Control Turn Off In (White arrow block)
    - Message in from Manual Override Control to turn off 1 hour switch
- 1-Hour Switch State
    - Check if 1-hour switch is on, if so then the message will go through to shut it off when the manual override control is turned on.
- Manual Control Turned On
    - Sets msg.payload to false and msg.manualControlTurnOff to true.

Manual Override Control



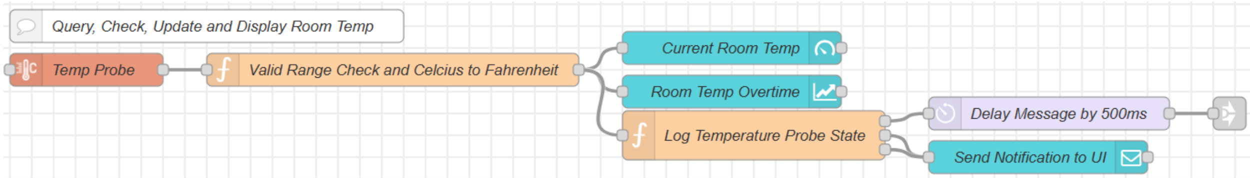
Overview – This flow controls the manual override control switch to forcefully turn the heater on nonstop for two hours before automatically turning off.

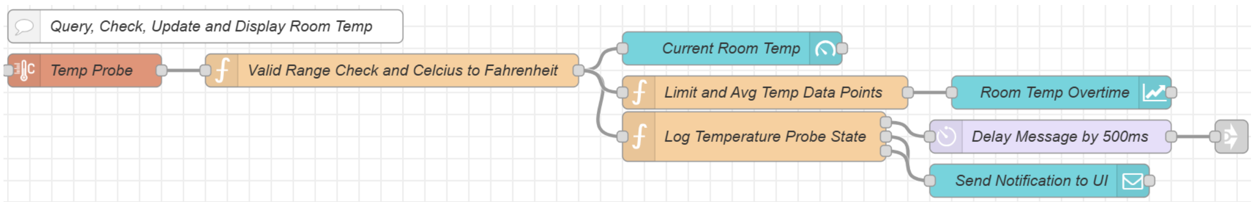
Node Details

- Heater Switch Control In (White arrow block)
    - Receives message from Heater Switch Control Out.
- Manual Override Control
    - UI switch for manual override control.
- State
    - Check if manual override control switch has been turned on or off.

- Disable
    - Sets “manualControlState” global variable to false, “currentHeaterPinState“ global variable to false, msg.payload to false, and msg.reset to blank to reset the 2-hour timer countdown to shutoff the heater.
- Enable
    - Sets “manualControlState” global variable to true, “currentHeaterPinState” global variable to true and msg.payload to true.
- 2 Hour Timeout
    - Tracks a 2-hour timer to shut off manual override control switch and heater after two hours.
- 1-Hour Switch Control Turn Off Out (White arrow block)
    - Sends message to “1-Hour Switch Control Turn Off In” to potentially turn off the 1-hour switch if it was on.
- Adjust Payload
    - Sets msg.payload to false to turn the manual override control switch and heater off.
- LED Indicator Manual Override Out (White arrow block)
    - Links to ui LED indicator to update.
- Log Manual Control
    - Creates log message when the manual override control switch is turned on or off.
- Heater Relay Switch
    - GPIO 18 pin that is turned on or off by the manual override control switch.
- Heater Switch Control Out (White arrow block)
    - Passes 2-hour timeout shut off message to “Heater Switch Control In” to update ui.
- Write To Heater Log In (White arrow block)
    - Routes all HeaterState.log messages to be written in next node.
- Filename and Path
    - Set filename path for HeaterState.log.
- Write Heater Log
    - Writes by appending to HeaterState.log.

Query, Check, Update and Display Room Temp





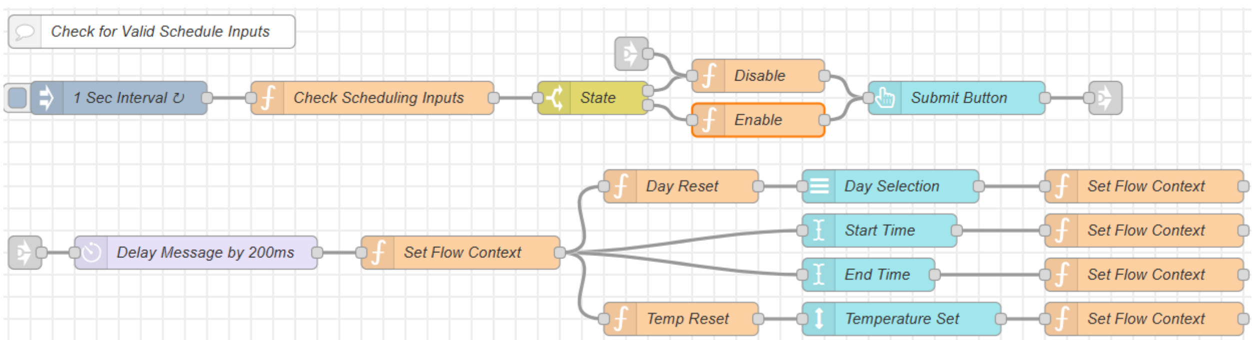
Overview – This flow gets the temperature probe reading then verifies its within the data sheets range of -55C to 125C. Once that is good it will convert the temperature from Celsius to Fahrenheit and shows this on the temperature gauge and a graph. With this monitoring the temperature range and temperature probe connection or disconnection, if either happens a notification is sent to the UI.

Node Details

- Temp Probe
  - Get temperature from DS18B20 temperature sensor.
- Valid Range Check and Celcius to Fahrenheit
  - This node checks that the temperature probe is connected and will send the proper errors if not. Then also checks that the temperature reading is within its rated range of -55C to 125C and will send the proper errors if not. If either of these errors are detected the heater will be shutoff. Note with detecting when the probe disconnects is it can take and average of three minutes for it to detect that the temperature probe has disconnected this is due to the 1 wire bus taking time to update on the Raspberry Pi.
- Current Room Temp
  - UI gauge showing real time room temperature.
- Limit and Avg Temp Data Points
  - Averages the temperature every minute to send to the Room Temp Overtime node.
- Room Temp Overtime
  - UI graph showing room temp over the last 12 hours you can change this in the node by changing the X-Axis Limit time from 12 hours to what you desire.
- Log Temperature Probe State

- Logs if temperature probe becomes connected, disconnected, out of range or back within range. This will send the downstream message to notify the UI of any issues such as disconnection or out of range.
- Delay Message 500ms
  - Message waits half a second to continue through.
- Send Notification to UI
  - Displays notification to UI.
- Log Temperature Sensor State Out (White arrow block)
  - Route log message to “Write To Heater Log In” to write to HeaterState.log.

Check for Valid Schedule Inputs



Overview – This flow looks for the user adding a new time period to the schedule and will check for the correct conditions to allow the information to be submitted.

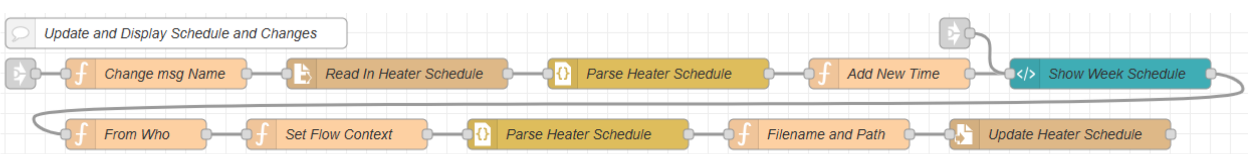
Node Details

- 1 Sec Interval
  - Injects message every second to check for valid period information to be potentially added.
- Checking Scheduling Inputs
  - Checks day selection, start time, end time and temperature for valid values to enable submit button.
- State
  - Check if ready to enable submit button.
- Init in 2 (White arrow block)
  - Ensures the submit button is disabled when system boots.
- Disable
  - Sets msg.enabled to false making button disabled.
- Enable

- Sets msg.enabled to true making button enabled.

- Submit Button
  - UI button that when pressed starts the process to add the time period and temperature to the HeaterSchedule.json file.
- Update Schedule Out (White arrow block)
  - Routes messages to “Update Schedule In 1” and “Update Schedule In 2”.
- Update Schedule In 1 (White arrow block)
  - Receives message from “Update Schedule Out”.
- Delay Message by 200ms
  - Delays message to not clear inputs instantly.
- Set Flow Context
  - Sets day selection, start time, end time and temperature to null effectively resetting the values and msg.payload to empty.
- Day Reset
  - Sets msg.payload to an empty array and msg.ui_update to null to reset input box.
- Day Selection
  - UI dropdown to select day.
- Start Time
  - UI text input to set start time of period.
- End Time
  - UI text input to set end time of period.
- Temp Reset
  - Sets msg.payload to an empty array and msg.ui_update to null to reset input box.
- Temperature Set
  - UI number input to set temperature for period.
- Set Flow Context (x4)
  - Sets appropriate flow variables for day selection, start time, end time and temperature.

Update and Display Schedule and Changes

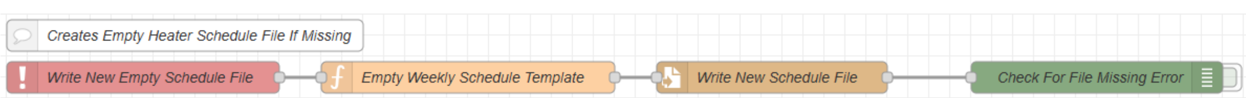


Overview – This flow works with the prior flow to add the period to the schedule.

Node Details

- Update Schedule In 2 (White arrow block)
  - Receives message from "Update Schedule Out".
- Change msg Name
  - Assigns msg.newTime with the current payload and adds msg.filename for HeaterSchedule.json path.
- Read In Heater Schedule
  - Reads in HeaterSchedule.json file.
- Parse Heater Schedule
  - Parses heater schedule from JSON to JavaScript object.
- Add New Time
  - Adds new time to JavaScript object.
- Init in 1 (White arrow block)
  - Receives message from "Init out" to show the week schedule in the ui.
- Show Week Schedule
  - UI template node is used to build heater schedule display and to delete schedules.
- From Who
  - Blocks messages from going farther if from initialization.
- Set Flow Context
  - Sets global heaterSchedule to new updated schedule.
- Parse Heater Schedule
  - Converts JavaScript object back to JSON.
- Filename and Path
  - Sets filename path for HeaterSchedule.json.
- Update Heater Schedule
  - Writes new updated heater schedule to HeaterSchedule.json.

Creates Empty Heater Schedule File If Missing

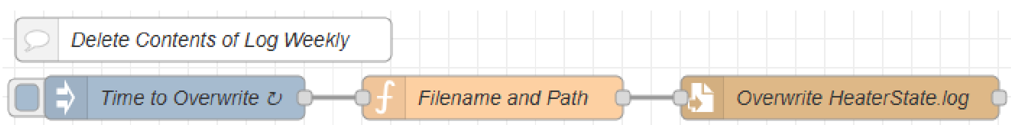


Overview – This flow ensures that if there is no HeaterSchedule.json it will write a new a blank template.

Node Details

- Write New Empty Schedule File
    - Node will catch errors when trying to read HeaterSchedule.json file if it doesn't exist.
- Empty Weekly Schedule Template
    - Empty JavaScript template object for schedule and includes the filename and path to write to.
- Write New Schedule File
    - Write the new HeaterSchedule.json file.
- Check For File Missing Error
    - Used to output debug message if this flow was activated.

Delete Contents of Log Weekly

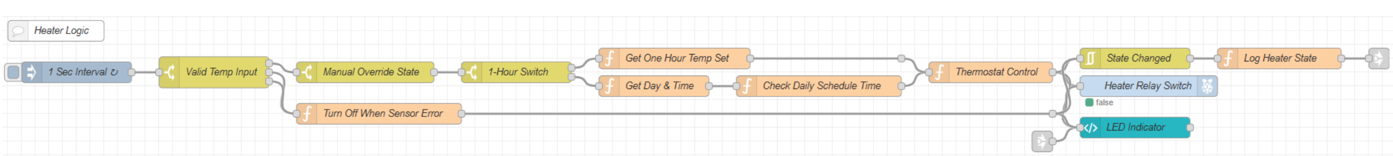


Overview – This flow will overwrite the HeaterState.log file every week by default is set to 5am on Sunday morning.

Node Details

- Time to Overwrite
    - Send a message at the time to overwrite the HeaterState.log file to clear. The default time can be changed by selecting this node then deploying the new change.
- Filename and Path
    - Set the filename path for HeaterState.log.
- Overwrite HeaterState.log
    - Overwrites the file to make it empty.

Heater Logic Flows



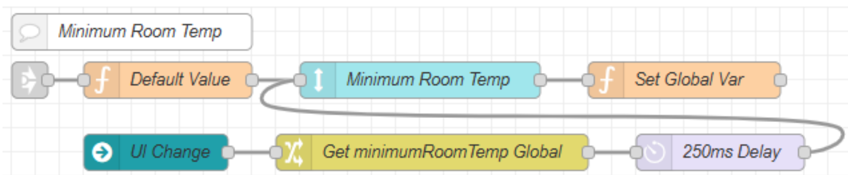
Overview – This flow is the actual logic to determine when to change the heater state (GPIO 18 pin).

Node Details

- 1 Sec Interval
  - After a two second delay it injects a message every second to check heater logic.
- Valid Temp Input
  - Checks that tempProbeError global is valid by ensuring it's not true or undefined.
- Manual Override State
  - Checks that the manual override state is off.
- Turn Off When Sensor Error
  - Turns off heater when no temperature probe is detected or out of temperature range.
- 1-Hour Switch
  - Checks that the 1-Hour Switch is on or off.
- Get One Hour Temp Set
  - Retrieves the global oneHourTempSet.
- Get Date & Time
  - Gets the current date and time to make msg.day, msg.hour and msg.min.
- Check Daily Schedule Time
  - Iterates through global heaterSchedule object to find if any periods should be running currently.
- Thermostat Control
  - Accounts for several global variables to find the highest setpoint and may change the state of the heater depending on the room temperature.
- LED Indicator Manual Override In (White arrow block)
  - Receives messages from entire program to update LED indicator.
- State Changed
  - Checks for when msg.payload changes to record log to HeaterState.log.
- Heater Relay Switch
  - GPIO 18 pin that is turned on or off by thermostat control node.
- LED Indicator
  - Indicates heater state by displaying "Heater On" and turning green then off by displaying "Heater Off" and turning red.

- Log Heater State
  - Creates log message to write to HeaterState.log.
- Log Schedule Control Out (White arrow block)
  - Routes log message to “Write To Heater Log In” to write to HeaterState.log.

Min Room Temp Flows



Overview – This flow sets the default valve of the Minimum Room Temp and updates a global variable.

Node Details

- Min Room Temp In (White arrow block)
  - Gets message from “Min Room Temp Out” and routes to next node
- Default Value
  - Holds integer for default value for minimum room temp which is 50°F. You can change this by selecting the node and change the value of 50 to your desired temperature then deploy the change. I would not go higher than what you are comfortable heating your room too or make the heater run too often.
- Minimum Room Temp
  - Numeric user interface (ui) node that allows the user to change this value from the ui. The default range is 0°F to 80°F to change this select the node and change the values then deploy your change.
- Set Global Var
  - Sets global variable “minimumRoomTemp” when changed from ui.
- UI Change
  - Detects a refresh or page change happens to make sure that the oneHourTempSet value displayed is accurate.
- Get minimumRoomTemp Global
  - Get the stored value for the minimumRoomTemp.
- 250ms Delay
  - Delays message by 250ms.

Debug Flows

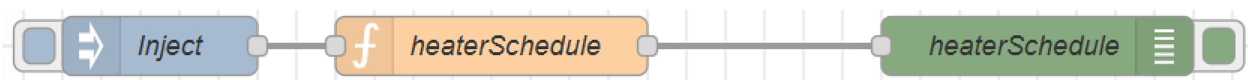
currentHeaterPinState

Overview – This flow allows you to poll the value of currentHeaterPinState at any time and read in the debug window in Node-RED.

Node Details

- Inject
  - Select to start message injection.
- currentHeaterPinState
  - Get the current value of currentHeaterPinState.
- currentHeaterPinState (green)
  - Displays value in debug window.

heaterSchedule



Overview – This flow allows you to poll the value of heaterSchedule at any time and read in the debug window in Node-RED.

Node Details

- Inject
  - Select to start message injection.
- heaterSchedule
  - Get the current value of heaterSchedule.
- heaterSchedule (green)
  - Displays value in debug window.

lastHeaterSchedule

Overview – This flow allows you to poll the value of lastHeaterSource at any time and read in the debug window in Node-RED.

Node Details

- Inject
  - Select to start message injection.
- lastHeaterSource
  - Get the current value of lastHeaterSource.
- lastHeaterSource (green)
  - Displays value in debug window.

manualControlState

Overview – This flow allows you to poll the value of manualControlState at any time and read in the debug window in Node-RED.

Node Details

- Inject
  - Select to start message injection.
- manualControlState
  - Get the current value of manualControlState.
- manualControlState (green)
  - Displays value in debug window.

minimumRoomTemp

Overview – This flow allows you to poll the value of minimumRoomTemp at any time and read in the debug window in Node-RED.

Node Details

- Inject
    - Select to start message injection.
- minimumRoomTemp
    - Get the current value of minimumRoomTemp.
- minimumRoomTemp (green)
    - Displays value in debug window.

oneHourSwitchState

Overview – This flow allows you to poll the value of oneHourSwitchState at any time and read in the debug window in Node-RED.

Node Details

- Inject
    - Select to start message injection.
- oneHourSwitchState
    - Get the current value of oneHourSwitchState.
- oneHourSwitchState (green)
    - Displays value in debug window.

oneHourTempSet

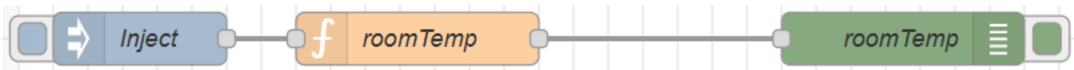
Overview – This flow allows you to poll the value of oneHourTempSet at any time and read in the debug window in Node-RED.

Node Details

- Inject
    - Select to start message injection.
- oneHourTempSet
    - Get the current value of oneHourTempSet.
- oneHourTempSet (green)

- Displays value in debug window.

roomTemp



Overview – This flow allows you to poll the value of roomTemp at any time and read in the debug window in Node-RED.

Node Details

- Inject
  - Select to start message injection.
- roomTemp
  - Get the current value of roomTemp.
- roomTemp
  - Displays value in debug window.

tempProbeError

Overview – This flow allows you to poll the value of tempProbeError at any time and read in the debug window in Node-RED.

Node Details

- Inject
  - Select to start message injection.
- tempProbeError
  - Get the current value of tempProbeError.
- tempProbeError
  - Displays value in debug window.